

COSC 50523: Deep Learning Course Project Description

Fall 2026

Purpose. The course project asks you to apply deep learning to a meaningful problem and demonstrate a complete experimental workflow. You must define a clear task, establish a credible baseline, evaluate a suitable neural model, and communicate the findings and limitations. The project is worth **30% of the final course grade**.

1 Project at a Glance

Component	Course Weight	Syllabus Timing
Team confirmation and project idea	5%	Team confirmation: September 2; project idea: September 16
Midterm progress report with implementation details	10%	October 14
Final report and implementation	10%	December 2
Final presentation	5%	December 7 and December 9
Bonus	5-8%	Evaluated after the final presentations.

The dates above follow the course syllabus. Submission links, exact submission times, presentation order, and any approved schedule changes will be posted on TCU Online.

1.1 Team Rules

- **Undergraduate students:** Work individually or in a team of no more than three students.
- **Graduate students:** Complete the project individually.
- Team membership is fixed after confirmation unless the instructor approves a change.
- For an undergraduate team, one representative should send the confirmation with the subject COSC-50523 Team [TeamName] and copy all team members.
- All team members are responsible for understanding the complete project, not only the part they implemented.

2 Selecting an Appropriate Project

Choose one of the following project paths. Every topic requires instructor approval.

2.1 Path A: Applied Deep Learning

Use deep learning to address a real problem in science, engineering, health, language, vision, security, or another domain. The task may involve classification, regression, detection, segmentation, forecasting, generation, representation learning, or multimodal learning.

2.2 Path B: Model Reproduction and Extension

Reproduce a clearly defined result from a recent paper or public implementation, then test a meaningful change. Examples include modifying an architecture, loss function, training strategy, data representation, or evaluation procedure. Reproduction by itself is insufficient; the project must include an extension or a new analysis.

2.3 Path C: Competition-Based Project

Use an active competition as the problem and data source. The project must still satisfy the course requirements. A leaderboard score alone is not a course project. You must include your own baseline, controlled experiments, reproducible implementation, and analysis of the results.

Scope check. The project must make deep learning technically necessary and relevant. A task that can be completed adequately with a simple classical model will normally be too limited unless the proposal identifies a clear deep learning research question. Projects that require unavailable data, excessive computation, or inaccessible proprietary services will not be approved.

2.4 Minimum Technical Expectations

Every project should include the following:

- a precise input, output, and learning objective;
- a documented dataset with legal and ethical permission for the proposed use;
- a defensible training, validation, and test protocol that avoids data leakage;
- at least one simple, credible baseline;
- at least one neural architecture relevant to the course, such as an MLP, CNN, autoencoder, VAE, GAN, attention model, transformer, or multimodal network;
- metrics appropriate to the task and class distribution;
- a controlled comparison, such as an ablation or architecture comparison;
- qualitative analysis when appropriate, including representative successes and failure cases; and
- enough documentation to reproduce the main result.

Graduate projects should show greater independence and technical depth. This normally means a stronger literature foundation, a meaningful model or experimental extension, and a more systematic evaluation.

3 Deliverables

Required report format. Beginning with the project idea, all written project submissions must use the [official IEEE Conference Template on Overleaf](#). Do not change the prescribed margins, font sizes, column widths, or line spacing.

3.1 Team Confirmation and Project Idea (5%)

The project idea should be a concise one-page proposal, excluding references. It should answer five questions:

1. What problem will you study, and why does it matter?
2. What dataset will you use, and can you access it now?
3. What neural approach will you test, and what will serve as the baseline?
4. How will you evaluate success?
5. Why is the project feasible within the semester and available computing resources?

Include the project title, student name or team name, all member names, dataset link, and two or more relevant references. Submit the proposal as a PDF through TCU Online. Approval may require a revision if the topic is too broad, too limited, or infeasible.

3.2 Midterm Progress Report with Implementation Details (10%)

The midterm report should be up to three pages, excluding references. It must show actual implementation progress rather than plans alone. Include:

- the updated problem statement and any change from the approved proposal;
- dataset statistics, preprocessing, split strategy, and checks for leakage;
- the baseline and proposed model, including enough architectural and training detail to understand the implementation;
- at least one completed experiment with a preliminary quantitative result if available;
- current difficulties, limitations, and a realistic plan for the remaining work; and
- a link to the working code or the requested code snapshot.

The report should use tables or figures when they communicate progress more clearly than prose. Results must identify the data split and metric being reported.

3.3 Final Report and Implementation (10%)

The final report must contain exactly five pages of main content. References and optional appendices are not included in this limit. All material required for grading must appear within the five-page main report. Use the following structure when it fits the project:

1. **Abstract:** State the problem, approach, and main result in no more than 250 words.
2. **Introduction:** Explain the problem, motivation, contribution, and principal findings.

3. **Related Work:** Position the project relative to relevant published work and baselines.
4. **Data:** Describe provenance, size, labels, preprocessing, splits, and important limitations.
5. **Methods:** Explain the baseline, proposed model, objective, training procedure, and major design choices.
6. **Experiments and Results:** Present the protocol, metrics, comparisons, and controlled analysis using readable tables or figures.
7. **Discussion and Conclusion:** Interpret the results, analyze failures and limitations, and identify a defensible next step.

The implementation must contain a clear README, dependency information, instructions for running the main experiment, and the code needed to reproduce the reported result. Fix random seeds where practical and record the final configuration. Do not include large datasets or model checkpoints unless specifically requested; provide stable access instructions instead.

3.4 Final Presentation (5%)

The presentation should communicate the project as a coherent technical argument, not as a chronological list of tasks. The exact time limit and presentation order will be announced after the number of projects is finalized. **The presentation template slides will be provided by the instructor.**

A strong presentation should cover:

- the problem, motivation, and central question;
- the data and evaluation protocol;
- the baseline and proposed method, preferably with a clear architecture or pipeline figure;
- the most important quantitative comparison;
- one useful qualitative result or failure case;
- the main conclusion, limitations, and next step; and
- direct, technically accurate answers during questions.

All undergraduate team members must participate meaningfully and should be prepared to answer questions about any part of the project. Slides must be readable from the back of the classroom. Avoid dense paragraphs, raw code, and tables that are too small to read.

4 How the Work Will Be Evaluated

Area	Evidence Expected
Problem and scope	A precise, important, and feasible deep learning question
Technical quality	Appropriate model choices, correct implementation, and justified design decisions
Experimental quality	Sound splits, suitable metrics, credible baseline, controlled comparisons, and no leakage
Analysis	Honest interpretation, failure analysis, limitations, and conclusions supported by evidence
Reproducibility	Organized code, recorded configuration, clear instructions, and results that can be traced to an experiment
Communication	Clear writing, effective figures and tables, proper citations, and a focused presentation

Complexity alone does not earn credit. A smaller model evaluated carefully is stronger than a large model used without a clear experimental question.

5 Optional Bonus Credit (up to 5%)

Items 1–4 below are worth **1.25% each**, for a maximum of **5%**. An eligible competition project may earn an additional **3%**. The maximum available bonus is **8%**.

1. **Actively maintained code repository:** Maintain a project repository with regular commits, including at least one substantive commit each week from every team member.
2. **Experiment tracking:** Use Weights & Biases (W&B) or a comparable tool (MLflow, Comet.ml, and ClearML) to record configurations, training curves, and experimental results. Use such service that provides free or academic plan. Students are not expected to pay for an experiment-tracking service.
3. **Model interpretation:** Include an appropriate interpretability or representation visualization, such as Grad-CAM, attention maps, or an embedding visualization, and explain what it shows.
4. **Runnable demonstration:** Provide a simple inference demonstration or interface using a notebook, Gradio, Streamlit, or a comparable tool.
5. **Additional Bonus (Only for Competition):** A competition-based project that finishes among the top five teams on the official final leaderboard will receive an additional 3% of the total grade.

6 Implementation, Attribution, and Academic Conduct

- You may use public papers, datasets, pretrained models, libraries, and implementations when their licenses permit it and when every source is cited clearly in both the report and code.
- Clearly distinguish code you wrote from code you adapted. Copying another student's or team's code is prohibited.
- Use of AI tools must follow the course syllabus and any additional instructor guidance. Any permitted use must be acknowledged. The evaluation, analysis, interpretation, synthesis, and final explanation must be your own work.
- You are responsible for verifying all generated text, code, citations, and results. Fabricated citations or unverified claims are academic and technical failures.

- Do not use private, sensitive, or restricted data without documented authorization. Medical, biometric, and human-subject data require particular care.
- Code must run. Before submission, start from a clean environment when possible and execute the complete evaluation path described in the README.

7 Finding a Topic or Dataset

Useful starting points include [Kaggle Competitions](#), [DrivenData Competitions](#), [Zindi Competitions](#), [Hugging Face Datasets](#), [Papers with Code Datasets](#), and [Google Dataset Search](#).

Before committing to a dataset, verify its size, label quality, license, download access, compute requirements, and compatibility with the course timeline.

7.1 Popular Computer Vision Dataset Collections and Benchmarks

The following resources may help you identify a computer vision problem or benchmark. Dataset licenses and access conditions must be checked before use.

- **CVPapers Dataset Collection:** A broad collection of computer vision datasets.
- **YACVID Dataset Index:** An additional index of public computer vision datasets.
- **ImageNet:** A large-scale image database organized according to the WordNet hierarchy.
- **SUN Database:** A benchmark for scene recognition and object detection with annotated scene categories and segmented objects.
- **Places Database:** A scene-centric dataset; Places205 contains 205 scene categories and approximately 2.5 million labeled images.
- **NYU Depth Dataset V2:** Aligned RGB and depth observations from labeled indoor scenes.
- **Microsoft COCO:** A benchmark for object recognition, detection, segmentation, and image captioning.
- **Labeled Faces in the Wild:** More than 13,000 labeled face photographs for unconstrained face recognition.
- **MPII Human Pose Dataset:** A benchmark for articulated human pose estimation.
- **YouTube Faces Database:** Face videos for unconstrained face recognition in video.
- **UCF101:** Realistic action videos organized into 101 action classes.
- **HMDB51:** A human-motion dataset organized into 51 action classes.
- **ActivityNet:** A large-scale benchmark for human activity understanding in video.
- **Moments in Time:** A dataset of approximately one million three-second video clips.

7.2 Examples of Active Competitions

The following competitions were active when this guide was prepared on September 9, 2026. Competition dates and rules can change, so verify the official page before proposing one.

- **Solar Filament Segmentation Challenge 2026** (deadline: November 15). Scientific image segmentation; a strong fit for CNN, U-Net, or vision-transformer experiments.
- **RSNA Knee Abnormality Detection** (deadline: October 22). Multimodal MRI and report modeling; a strong but computationally demanding option that requires careful data handling.
- **Benchmarking WSI Privacy** (deadline: October 31). Whole-slide image retrieval and privacy; an advanced option with substantial data and compute demands.
- **ARC Prize 2026: ARC-AGI-3** (deadline: November 2). Interactive reasoning agents; a highly advanced, nontraditional option that requires a narrow scope and prior approval.